



Protokoll 1

$$\textcircled{1} U = I \cdot R$$

$$\textcircled{2} I_{\text{ges}} = I_1 + I_2$$

$$I_{\text{ges}} = \frac{U_1}{R_1} + \frac{U_2}{R_2}$$

$$U_1 = I_1 \cdot R_1 \Rightarrow \frac{U_1}{R_1} = I_1$$

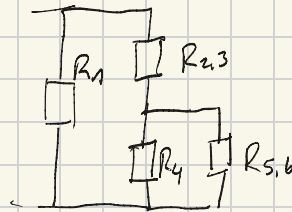
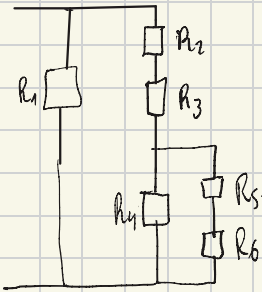
$$R_{2,3} = R_2 + R_3 = 1000 + 47 \Omega$$

$$R_{2,3} = 1047 \Omega$$

$$R_{5,6} = R_5 + R_6 = (330 + 220) \Omega$$

$$R_{5,6} = 550 \Omega$$

Reihenfolge: $R_{\text{ges}} = \sum R_n$
 Parallel: $R_{\text{ges}} = \frac{R_1 \cdot R_2}{R_1 + R_2}$



$$2) R_4 \parallel R_{5,6} \quad R_{4,5,6} = \frac{R_4 \cdot R_{5,6}}{R_4 + R_{5,6}} = \frac{4700 \Omega \cdot 550 \Omega}{4700 \Omega + 550 \Omega} =$$

$$R_{4,5,6} = 492,4 \Omega$$

$$3) R_{2,3,4,5,6} = R_{2,3} + R_{4,5,6} = 1047 \Omega + 492,4 \Omega$$

$$R_{2,3,4,5,6} = 1539,4 \Omega$$

$$4) R_1 \parallel R_{23456} \quad R_{\text{ges}} = \frac{R_1 \cdot R_{23456}}{R_1 + R_{23456}} = \frac{470 \Omega \cdot 1539,4 \Omega}{470 \Omega + 1539,4 \Omega}$$

$$R_{\text{ges}} = 360,06 \Omega$$

$$5) I_{\text{ges}} = \frac{U_{\text{ges}}}{R_{\text{ges}}} \Rightarrow I_{\text{ges}} = \frac{10V}{360,06 \Omega} = 27,8 \text{ mA}$$

$$6) U_{\text{ges}} \parallel U_1, U_{2-6} = 10V$$

$$I_1 = \frac{U_1}{R_1} = \frac{10V}{470 \Omega} = 21,27 \text{ mA}$$

$$7) \quad I_{ges} = I_1 + I_2 \Rightarrow I_2 = I_{ges} - I_1 = 27,8 \text{ mA} - 21,27 \text{ mA} \\ I_2 = 6,53 \text{ mA} \\ I_2 = I_3$$

$$8) \quad U_2 = I_2 \cdot R_2 = 6,53 \text{ mA} \cdot 1000 \Omega = 6,53 \text{ V} \\ U_3 = I_3 \cdot R_3 = 6,53 \text{ mA} \cdot 47 \Omega = 307 \text{ mV}$$

$$9) \quad U_2 + U_3 + U_4 = 10 \text{ V} \Rightarrow U_4 = 10 \text{ V} - U_2 - U_3 \\ U_4 = 10 \text{ V} - 6,53 \text{ V} - 0,307 \text{ V} = 3,163 \text{ V}$$

$$10) \quad I_4 = \frac{U_4}{R_4} = \frac{3,163 \text{ V}}{475 \Omega} = 0,673 \text{ mA}$$

$$11) \quad I_5 = \frac{U_{5,6}}{R_{5,6}} = \frac{3,163 \text{ V}}{550 \Omega} = 5,75 \text{ mA}$$

$$12) \quad U_5 = R_5 \cdot I_5 = 330 \Omega \cdot 5,75 \text{ mA} \approx 1,9 \text{ V}$$

$$13) \quad U_6 = R_6 \cdot I_6 = 220 \Omega \cdot 5,75 \text{ mA} = 1,265 \text{ V}$$